Quiz 1 — Last-Minute Revision

Built from the class notes and the lecture slide deck, in the order they were taught. J. M. Smith (*Chemical Engineering Kinetics*, Ch. 7–8) is used only where the notes are compressed — those additions are boxed and labelled, so you can always tell lecture material from book material.

Anchors: **CRE Quiz 1 notes (21 pp.)** · **CRE-II_Q1 lecture notes / ppt (10 slides)** · Ref: JMS Ch. 7, Ch. 8-1 → 8-10

IF YOU ONLY HAVE 20 MINUTES

§18 formula sheet → §19 numbers → the **seven steps** (§5) → the **physisorption vs chemisorption** table (§10) → the **Langmuir derivation** (§11) → the **five densities** (§15). Everything else is recall-only.

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§ 1 · Lecture 2-3 revision

Where this course starts

Designing a reactor = **kinetics** + **flow pattern**. CRE-I gave you both for a single phase. CRE-II keeps the same design equations and changes only what goes into the rate.

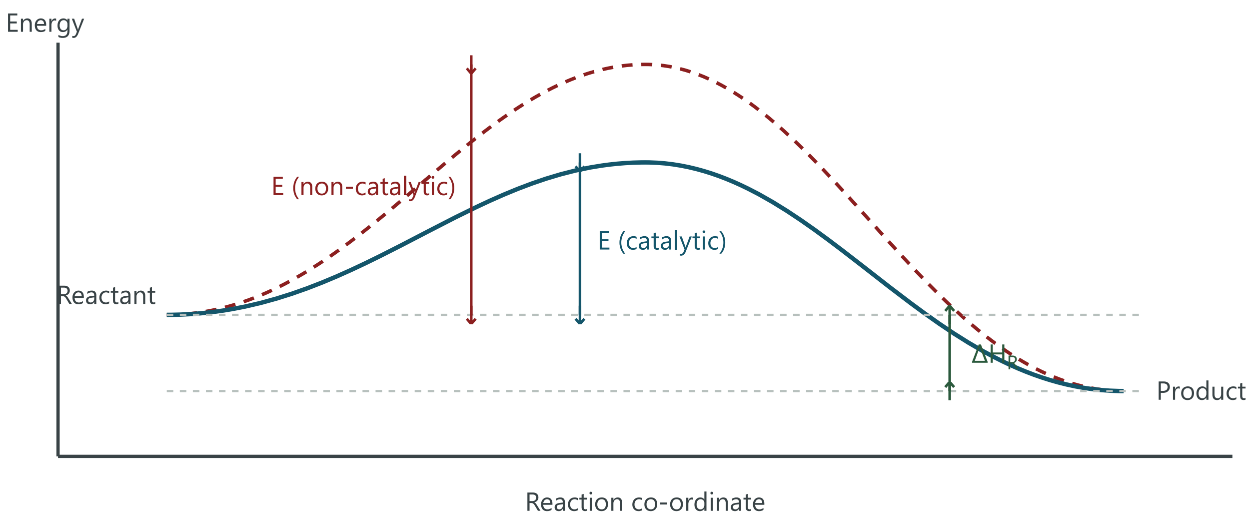
KINETICS — THE POWER LAW

$$(-r_A) = k\,C_A{}^n \;=\; k_0\,e^{-E_a/RT}\,C_A{}^n$$

FLOW PATTERN — THE TWO IDEAL LIMITS

- **Complete mixing** → CSTR / MFR
- **No axial mixing** → PFR

▸ The energy diagram — what a catalyst actually changes



The catalyst lowers E_a only. The two levels — and therefore ΔH_R — do not move.

THE SINGLE MOST EXAMINABLE SENTENCE IN MODULE 1

A catalyst **alters the rate but not the thermodynamics**. ΔH, ΔG and the equilibrium conversion are unchanged; only the activation energy changes. A catalyst that speeds the forward reaction speeds the reverse one by exactly the same factor.

Multi-step reactions and the rate-determining step

For a sequence **A → B → C** proceeding through an intermediate **A***:

$$\mathrm{A} \rightarrow \mathrm{A}^{\bullet} \ \text{ is the RDS } \ \text{when } \ E_{a1} > E_{a2}$$

The step with the **largest** activation-energy barrier is the slowest, and therefore controls the overall rate.

§ 2 · Lecture 5

Classifying heterogeneous reactions

A reaction is **heterogeneous** when more than one phase is present — so a phase boundary must be crossed before anything can react. Two cuts are made in the notes: first by **phases involved**, then by **catalytic vs non-catalytic**.

Cut 1 — by phases

Gas + solid · Liquid + solid · Liquid + liquid · Solid + solid

Cut 2 — catalytic or not

CLASS NOTES, PAGE 2 — WITH THE PHASES EACH ONE RUNS IN

Class	Examples given in lecture	Phases
Non-catalytic	Combustion of a solid; gasification; partial combustion	G/S — a solid *reactant* is consumed, so the rate changes with time (inherently transient)
Catalytic	Hydrocracking	G/S
	Catalytic cracking / reforming	
	Hydrogenation / dehydrogenation	
	Desulphurisation (HDS)	G/S/L
	Alkylation	Little solid / mostly liquid

JMS 7-2 — THE TWO ROWS THE LECTURE DIDN'T SPELL OUT

- **Gas–liquid–solid:** slurry hydrogenation, ethylene polymerisation, trickle beds. Several physical steps in series; diffusion of dissolved H_2 to the particle is usually the slow one.
- **Solid–solid:** ceramics manufacture. Interface kinetics are essentially unknown because every measurement already contains diffusion.
- Non-catalytic G/S examples worth naming: UO_2 + 4HF, roasting ZnS, $CaCO_3 \rightarrow CaO$ in a lime kiln, and burning coke off a spent cracking catalyst.

§ 3 · Lecture 6

Defining the rate — three bases

For a homogeneous reaction the rate per unit volume is the only sensible basis. With a solid catalyst in the vessel, the same disappearance of A can be normalised three different ways — and the quiz will expect you to convert between them.

$$-r_V = \frac{1}{V}\,\frac{dN}{dt}$$

mol / (m³ reactor · s)

$$-r_W = \frac{1}{W}\,\frac{dN}{dt}$$

mol / (kg cat · s)

$$-r_S = \frac{1}{S_A}\,\frac{dN}{dt}$$

mol / (m² cat · s)

$$(-r_V)\,V = (-r_W)\,W = (-r_S)\,S_A$$

The same number of moles per second, three ways of writing it. This is the bridge between a lab rate (per gram) and a design rate (per m³).

JMS 7 — THE CONVERSION THE DESIGN EQUATION ACTUALLY USES

$r_V = \rho_B\, r_W$, where ρ_B is the **bulk density of the packed bed**. Rates are measured per gram of catalyst; mole balances want them per unit reactor volume.

Also learn the pair of names: the **intrinsic (chemical) rate** covers steps 3–5 only and is written in terms of conditions at the surface; the **global (overall) rate** lumps all seven steps and is written in terms of the **bulk** fluid — because bulk T and C are the only things you can measure or specify.

§ 4 · Lecture 5–6

What a catalyst must do

Five requirements, straight from the notes. Expect a one-mark "state the properties of a catalyst" question.

1. **It alters the rate, not the thermodynamics.** ΔH and ΔG are unchanged — only E_a changes. The driving force of the reaction is untouched.
2. **It must provide a high surface area** — of order **10^2–10^3 m²/g**. Adsorption is a surface phenomenon, so the number of active sites scales with area.
3. **Supported or unsupported** (bulk). Unsupported example: **Fe for NH_3 synthesis** (fertiliser).
4. **Deposition of the active agent on a support** is what gives mechanical and thermal stability — typically **SiO_2** or **Al_2O_3**.
5. **It must create or contain favourable sites** — the geometry and electronics of the site decide both activity and selectivity.

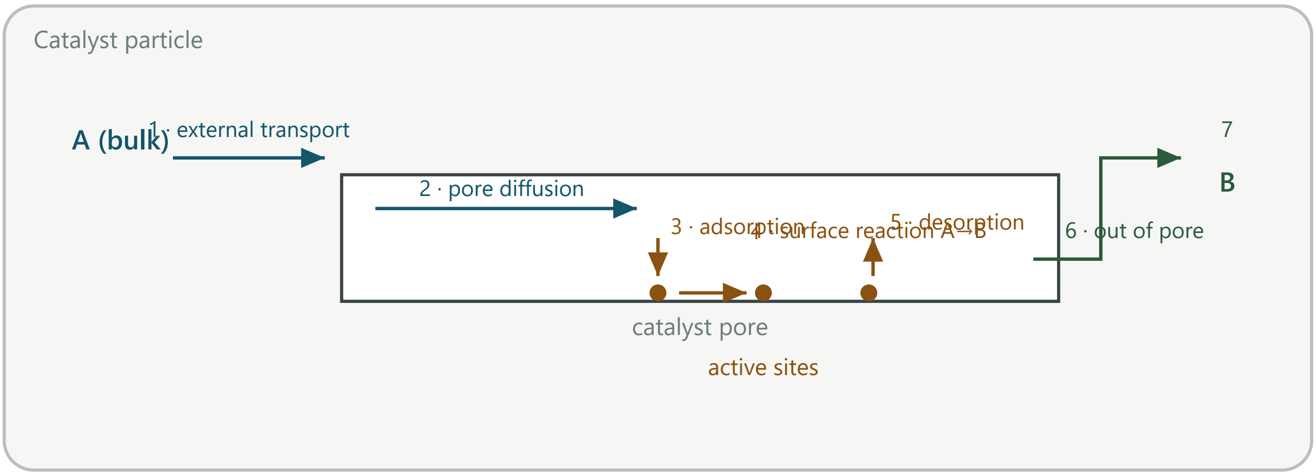
JMS 8-1 — EVIDENCE FOR EACH CLAIM, IF YOU WANT A FULLER ANSWER

- *Small quantity, large effect:* Cu^{2+} at 10^{-9} mol/L measurably speeds Na_2S oxidation.
- *Equilibrium unchanged:* SO_2 oxidation over Pt, Fe_2O_3 and V_2O_5 all give the **same** equilibrium composition.
- *Rate ∝ area:* for $H_2 + O_2$ on platinum the rate is directly proportional to the platinum surface.
- *Selectivity:* ethanol over different catalysts gives entirely different products — the classic selectivity example.

§ 5 · Lecture 11 · Slide 4

The seven steps of a catalytic reaction

This list is the skeleton of the whole course. Learn it **in order** and know which steps are physical and which are chemical.



Steps 1 & 7 external physical · 2 & 6 internal physical · 3, 4, 5 chemical (= the intrinsic rate).

THE SEVEN STEPS, WITH THE NOTE THAT GOES WITH EACH

#	Step	What the lecture attached to it
1	**Mass transfer** of A from the bulk to the outer catalyst surface	Governed by C_A and the mass-transfer coefficient
2	**Diffusion** of A into the pores to the active sites	Catalyst size ≈ **50 μm** → **300 μm**; pore size ≈ **Å** = 10^{-10} **m**. The pore mouth must be larger than the diffusing molecule, or the molecule never gets in.
3	**Adsorption** of A — it binds to an active site	A **surface phenomenon**: adsorption ∝ surface area
4	**Surface reaction** A → B	Adsorption and reaction can occur on *different* sites
5	**Desorption** of B	Frees the site so the cycle can repeat
6	**Diffusion** of B back to the outer surface	Mirror of step 2
7	**Mass transfer** of B to the bulk	Mirror of step 1

TWO CONSEQUENCES THE NOTES UNDERLINE

- **Every step has its own activation energy**, and $E_A \propto 1/\text{rate}$ — a large barrier means a slow step. The slowest step is the **rate-determining step**.
- ***k* governs the speed of the reaction** — the rate constant is what carries the temperature dependence into the rate expression.
- For a **non-porous** catalyst, steps 2 and 6 vanish and only five steps remain.

JMS 7-1 — THE DERIVATION THAT TURNS THE SEVEN STEPS INTO ONE EQUATION

Take A(g) → B(g) on a non-porous, isothermal catalyst, first order at the surface. Only three steps survive, in series. At steady state their rates are *equal*:

$$r_p = k_m a_m (C_b - C_s) \quad \text{and} \quad r_p = k a_m C_s \qquad \text{7-1, 7-2}$$

Eliminate the unmeasurable C_s:

$$\boxed{r_p = \frac{a_m}{1/k + 1/k_m} C_b} \qquad \text{7-4}$$

Chemical and physical steps behave as **two resistances in series**, driven by the bulk concentration alone.

Three regimes: $k \ll k_m$ → reaction control ($C_s \approx C_b$); $k_m \ll k$ → diffusion control ($C_s \approx 0$). Under diffusion control k has dropped out, so the **apparent activation energy collapses** — and because k grows exponentially with T while k_m barely moves, **raising T pushes any system towards diffusion control**.

§ 6 · Slide 5

Catalyst shapes

CLASS NOTES, PAGES 5–6 — MEMORISE ONE PLUS AND ONE MINUS FOR EACH

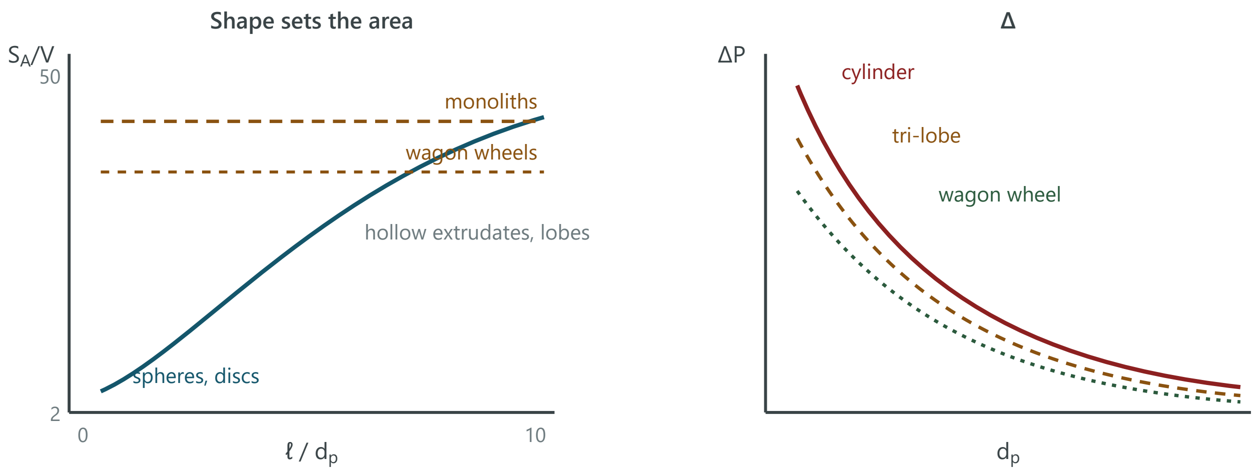
Shape	Characteristics	Used for
Spherical	Low manufacturing cost · high pressure drop · large diffusion length	HDS, methanation
Irregular granule	Low manufacturing cost · high pressure drop · low surface area	Ammonia, formaldehyde
Pellet	Regular shape · good strength · low surface area	CO shift, hydrogenation
Extruded cylinder	Low pressure drop · poor shape and strength	HDS
Monolith	Low pressure drop · small diffusion length · poor radial heat transfer	Exhaust-gas cleaning
Wagon wheel, hollow extrudate, tri-lobe, quadra-lobe	The "shaped" family — engineered specifically to raise external area per unit volume without raising ΔP	

MEMORY HOOK

Going down the list — **sphere** → **granule** → **pellet** → **cylinder** → **lobed** → **monolith** — pressure drop falls and surface-area-per-volume rises, but mechanical strength gets worse. Every shape is a point on that trade-off.

§ 7 · Slide 6

Effect of shape: area vs pressure drop



$P \propto 1/dp$

Left: shaped catalysts break out of the sphere/disc band. Right: shrinking the particle to gain area costs pressure drop.

THE TRADE-OFF, STATED AS THE LECTURE STATED IT

$d_p \propto S_A$ — smaller particles give more external area per unit volume, which raises the overall rate per kg of catalyst. But **$\Delta P \propto 1/d_p$**, and if ΔP rises you need a bigger compressor or pump. The chosen shape is the one that maximises S_A/V *subject to* acceptable crush strength and an affordable pressure drop.

▸ **Ergun equation — where the $1/d_p$ comes from**

$$\frac{\Delta P}{L} = 150\,\frac{\mu v(1-\varepsilon)^2}{d_p^{\,2}\,\varepsilon^3} + 1.75\,\frac{\rho v^2(1-\varepsilon)}{d_p\,\varepsilon^3}$$

Viscous (laminar) term + inertial (turbulent) term. Both carry d_p in the denominator, and both blow up as the void fraction ε falls.

§ 8 · Lecture 7

Catalyst composition

"Catalyst" as loaded into the reactor is a composite. The notes break it into four parts:

Component	Examples from the notes	Role
Catalytic agent	*Metals:* Fe, Cu, Pt, Pd, Rh *Semiconductors:* NiO, ZnO *Insulators:* Al_2O_3, SiO_2	The active substance — where the chemistry happens
Support / carrier	SiO_2, Al_2O_3, SiC	The base material. Gives high area from a small amount of an expensive agent, plus mechanical and thermal stability
Promoter	K_2O	Added in small amounts during manufacture; improves activity, selectivity, or life
Inhibitor	Cl (halogens)	The opposite — deliberately *lowers* activity to suppress an unwanted side reaction and raise selectivity

Worked example from the notes: V_2O_5/SiO_2 — SiO_2 is the support, V_2O_5 is the active agent. It runs $SO_2 \rightarrow SO_3$ in **H_2SO_4 production**. A catalyst may also be **unsupported (bulk)** — Fe for ammonia synthesis is the standard example.

JMS 8-10 — THE PROMOTER/INHIBITOR EXAMPLES WORTH QUOTING

- **Promoter:** iron for NH_3 synthesis with Al_2O_3 (also CaO, K_2O) added — it prevents loss of surface area by **sintering**, giving higher activity over a longer life.
- **Inhibitor:** silver on alumina for ethylene oxide. Silver also burns ethylene right through to CO_2; adding **halogen compounds** inhibits the complete oxidation and restores selectivity.
- **Inhibitor ≠ poison.** An inhibitor is added deliberately during manufacture; a poison arrives during operation.

§ 9 · Lecture 7 · Slide 7

Catalyst manufacture

The main route — precipitation

$$\text{Precipitation / gel} \rightarrow \text{Decantation / filtration} \rightarrow \text{Washing} \rightarrow \text{Drying} \rightarrow \text{Crushing} \rightarrow \text{Calcination} \rightarrow \text{Impregnation} \rightarrow \text{Activation}$$

Example reaction from the notes:

$$\mathrm{Al(NO_3)_3 + NH_4OH \rightarrow Al(OH)_3\downarrow + NH_4NO_3}$$

The alumina precursor $Al(OH)_3$ is filtered, washed, dried and calcined to Al_2O_3. **Washing is critical** — residual salts act as poisons.

Sol–gel method

- Prepare a **liquid suspension** of particles (~nm to µm) by **hydrolysis of a precursor salt**.
- **Condensation** of that sol leads to **gel formation**.

The notes flag this as a special case of precipitation — silica- and alumina-based catalysts suit it because their precipitates are colloidal.

▸ Impregnation — wet vs dry

Method	Solution used	Transport mechanism into the pore
Wet impregnation	**Excess** precursor salt solution — the support is soaked, then the excess is drained	Diffusion + adsorption
Dry impregnation (incipient wetness)	**Exact** amount of precursor salt solution — just enough to fill the pore volume	Diffusion + capillary flow

In both cases the active metal ends up dispersed as small particles on the internal surface of the support.

> JMS 8-9 — THE DISPERSION RESULT THAT MAKES A GOOD SHORT ANSWER
>
> For ortho → para hydrogen conversion over NiO/alumina, a **5.0 wt% catalyst was *less* active than a 0.5 wt%** one, because the concentrated soaking solution deposited larger nickel particles. Building the same 5.0 wt% up as **ten separate 0.5 wt% depositions gave 11× the activity** — same nickel, far better dispersion. Moral: beyond a small fraction of a monolayer, more loading does not help; **dispersion** does. That is why chemisorption area is measured rather than weight percent trusted.

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§ 10 · Lecture 11

Adsorption: physical vs chemical

Vocabulary first, because the quiz will use it precisely:

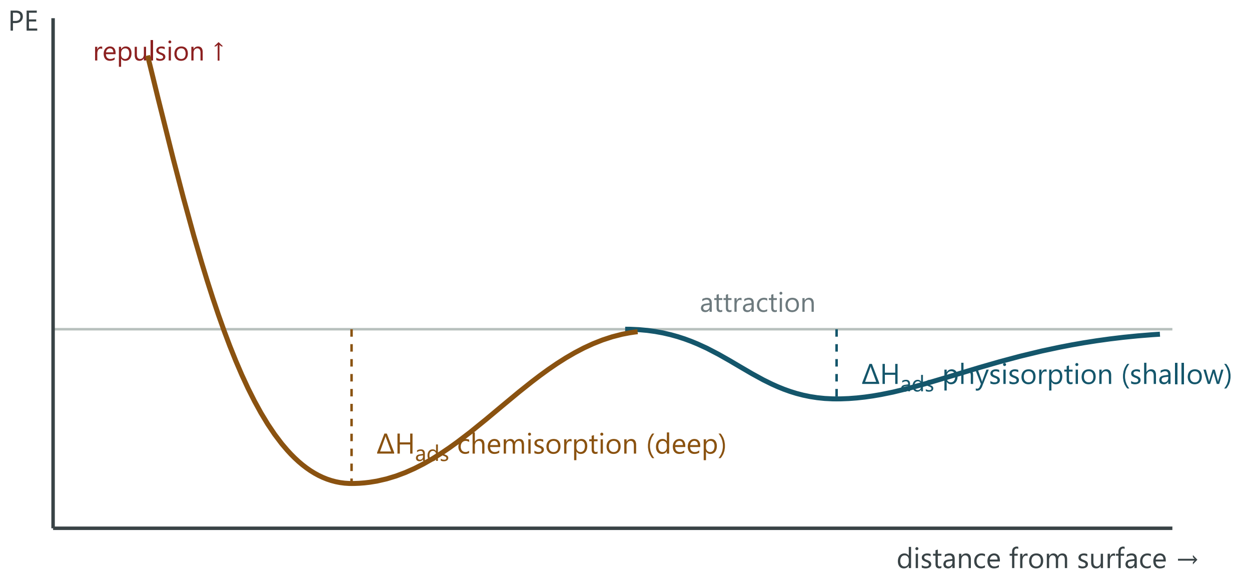
- **Adsorbate** — the molecule that sticks.
- **Adsorbent** — the surface it sticks to.

THE EIGHT-ROW COMPARISON — THE SINGLE MOST COMPRESSIBLE THING IN MODULE 1

	Physical adsorption (physisorption)	Chemical adsorption (chemisorption)
Forces	Van der Waals	Chemical bond
Surface required	**Any** surface can physisorb	Must be **chemically active**
Heat of adsorption	Low — **0.5 to 5 kcal/g-mol**	High — **5 to 100 kcal/g-mol**
Comparable to	~ ΔH of **condensation**	~ ΔH of **bond energy**
Temperature	Active at **low** T	Active at **high** T (relevant to catalysis, > 500 °C)
Reversibility	**Reversible**	**Irreversible**
Layers	Multilayer possible	Monolayer only
What it measures	**Surface area + pore volume**	**Number of active sites**

> SIGN CONVENTION — ALWAYS TRUE
>
> **Adsorption is exothermic:** $\Delta H_{ads} < 0$. A molecule that sticks has lost freedom, so entropy falls; for adsorption to be spontaneous ($\Delta G < 0$) the enthalpy term must be negative.

▸ The potential-energy curve



Physisorption is a shallow well far from the surface; chemisorption is a deep well at bond distance. The bottom of each well is ΔH_{ads}.

> **WHY BOTH MATTER FOR CHARACTERISATION**
>
> **Physisorption** (N_2 at −196 °C) → gives you **surface area and pore volume**, because it will cover *any* surface. **Chemisorption** (H_2, CO) → gives you the **number of active sites**, because it only binds where real bonding is possible. Two different experiments answering two different questions.

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§ 11 · Lecture 12

Adsorption isotherms & the Langmuir treatment

An adsorption isotherm is a plot of the quantity of **adsorbate** held on an **adsorbent**, at equilibrium, at a fixed temperature, against pressure (or concentration in a liquid). Equilibrium here means:

$$r_{\text{ads}} = r_{\text{des}}$$

rate at which molecules bind to the surface = rate at which they detach

▸ The four Langmuir assumptions

1. The **adsorbent surface is uniform**, and adsorbed molecules **do not interact** with each other.
2. The **entire surface is equally active** — every site is the same.
3. The **heat of adsorption is constant and independent of coverage** θ (the fraction of occupied sites).
4. Adsorption is limited to a **monolayer** — less than or equal to one layer.

▸ The derivation — know it line by line

Adsorption needs a collision with a **vacant** site, so it is proportional to pressure × the free fraction. Desorption only needs an occupied site:

$$r_{\text{ads}} = k\,p(1-\theta) \quad \cdot \quad r_{\text{des}} = k'\theta \qquad \text{8-1, 8-2}$$

At equilibrium set them equal and solve for θ:

$$k\,p(1-\theta) = k'\theta \implies k\,p = \theta(k' + k\,p) \implies \theta = \frac{k\,p}{k' + k\,p}$$

Divide through by k' and define the **adsorption equilibrium constant** $K = k/k'$:

$$\theta = \frac{K\,p}{1 + K\,p} = \frac{v}{v_{\mathrm{m}}} \qquad \text{8–3}$$

v = volume adsorbed · v_{m} = volume needed for a complete monolayer

The two limits

LOW PRESSURE — $KP \ll 1$

$\theta \approx Kp$. The isotherm is **linear with slope K** — Henry's-law region.

HIGH PRESSURE — $KP \gg 1$

$\theta \to 1$. The surface **saturates** at one monolayer; $v \to v_{\mathrm{m}}$.

Linearising it — two equivalent forms

Both put the Langmuir isotherm into `y = mx + c` form so that v_{m} and K come out of a straight line. Start from $v_{\mathrm{m}}/v = 1/(Kp) + 1$:

$$\frac{1}{v} = \frac{1}{K\,v_{\mathrm{m}}} \cdot \frac{1}{p} + \frac{1}{v_{\mathrm{m}}}$$

Plot $1/v$ vs $1/p$ · slope = $1/(K\,v_{\mathrm{m}})$ · intercept = $1/v_{\mathrm{m}}$

$$\frac{p}{v} = \frac{1}{K\,v_{\mathrm{m}}} + \frac{p}{v_{\mathrm{m}}} \qquad \text{8–8}$$

Plot p/v vs p · slope = $1/v_{\mathrm{m}}$ · intercept = $1/(K\,v_{\mathrm{m}})$. This is the form BET starts from.

WATCH THE ALGEBRA UNDER EXAM PRESSURE

Whichever linearisation you use, **get v_{m} first** (from whichever of slope/intercept contains it alone), then back out K from the other. Mixing up which is which is the standard way to lose the marks.

§ 12 · Lecture 12

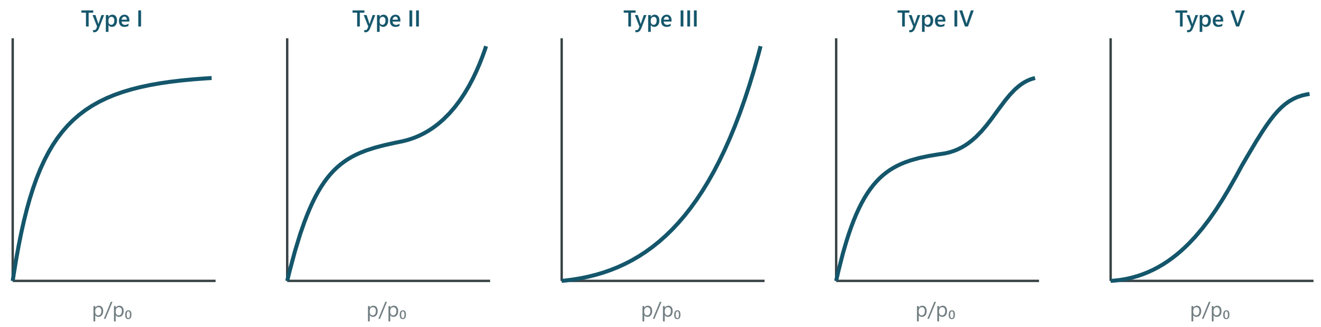
Temkin, Freundlich, and the IUPAC Types

Two isotherms that relax Langmuir's assumption 3

Langmuir assumed ΔH_{ads} is independent of coverage. In reality the best sites fill first, so **ΔH_{ads} falls as θ rises** — that is, $\Delta H_{\mathrm{ads}} \propto 1/\theta$. Two empirical fixes:

Isotherm	Coverage dependence of ΔH_{ads}	Resulting isotherm
Temkin	ΔH_{ads} falls **linearly** with θ	$\theta = k_1 \ln(k_2\, p)$ — k_1, k_2 constant at a given T
Freundlich	$\Delta H_{ads}(\theta) = \Delta H_0 \ln \theta$ — falls **logarithmically**	$\theta = c\, p^{1/n}$, with $n > 1$

The five IUPAC isotherm types



*y-axis is amount adsorbed; x-axis is p/p_0, where p_0 is the **vapour pressure of the adsorbate**.*

Type	Shape	What it means physically
I	Rises then plateaus	**Monolayer** — follows the Langmuir pattern. Microporous solids.
II	S-shaped, no plateau	**Multilayer.** First layer chemisorption, everything after that physisorption. Non-porous / macroporous.
III	Convex upward throughout	**Weak adsorbate–adsorbent interaction** — the molecule prefers its own kind to the surface.
IV	Type II, then a plateau	Multilayer + **capillary condensation** in mesopores. Shows a hysteresis loop.
V	Type III, then a plateau	**Weak adsorption + capillary condensation.**

§ 13 · Lecture 12

Pores and their sizes

Class	Diameter	Where it sits
Micropore	< 2 nm	Micropores are **inside** the catalyst pellet — that is where the surface area (and therefore the reaction) lives. Macropores are the **outside** channels that feed them.
Mesopore	2 nm < d_p < 50 nm	
Macropore	> 50 nm	

ONE-LINE SUMMARY FROM THE NOTES

"**Micropores are inside the catalyst; macro is outside.**" Transport happens in the macropores, reaction happens in the micropores — which is exactly why pore-size distribution, not just total area, decides catalyst performance.

§ 14 · Lecture 12 · Slide 8

BET — measuring surface area

Langmuir stops at a monolayer. Brunauer–Emmett–Teller extends it to **multilayer** adsorption, which is what N_2 actually does at −196 °C — and that extension is what makes the surface area measurable.

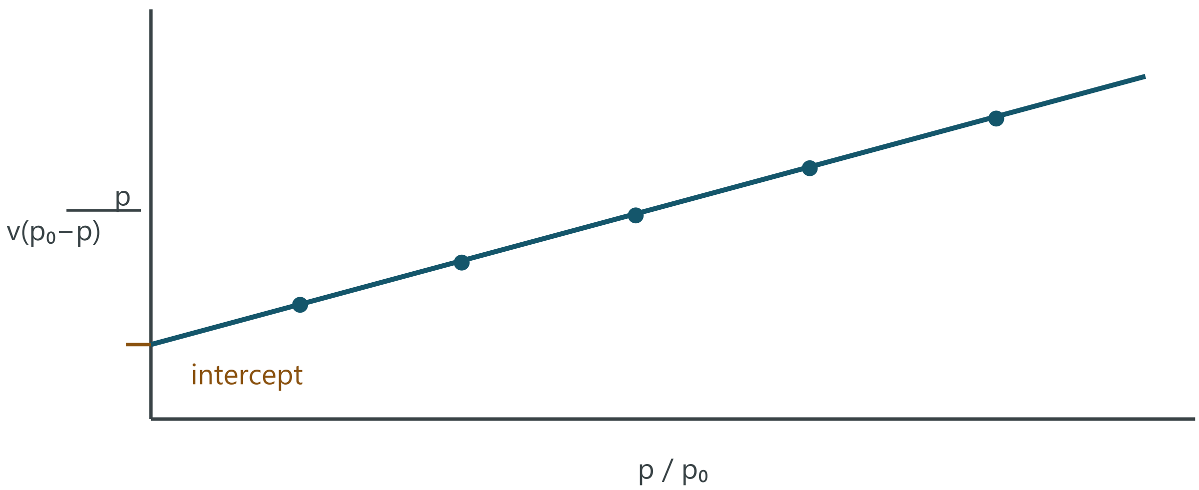
$$\frac{p}{v(p_0 - p)} = \frac{1}{v_m c} + \frac{c-1}{c\, v_m}\left(\frac{p}{p_0}\right) \tag{8-9}$$

c = BET constant · p_0 = vapour pressure of the adsorbate · v_m = monolayer volume

$$c = \exp\left(\frac{E_1 - E_L}{R\,T}\right)$$

E_1 = heat of adsorption of the first layer · E_L = heat of vaporisation (all later layers)

▸ Getting v_m and c off the plot



I = 1 / (vmc) slope s = (c−1) / (c vm)

$$v_m = \frac{1}{I + s} \qquad c = \frac{s}{I} + 1$$

8-10 → 8-12

Two unknowns, two pieces of information from one straight line. Add the intercept and slope — the c's cancel.

▸ From v_m to surface area

Convert the monolayer *volume* to a *number of molecules*, then multiply by the area each one covers:

$$\text{number of molecules in the monolayer} = \frac{N_0\, v_m}{V_{STP}}$$

N_0 = Avogadro's number · V_{STP} = 22,400 cm^3/mol

$$S_g = \frac{N_0\, v_m}{V_{STP}} \cdot \frac{\alpha}{m} \qquad \text{8-13}$$

α = area of one adsorbed molecule (m²) · m = mass of the sample · S_g = **specific** surface area, m²/g

JMS 8-5 — THE TWO SHORTCUTS

$\alpha = 1.09\,[M/(N_0\rho)]^{2/3}$ (8-14) gives the molecular area from liquid density. For **N_2 at −195.8 °C** this evaluates to **α = 16.2 Å²**, and the whole chain collapses to

$$S_g = 4.35 \times 10^4\, v_m \;\; \mathrm{cm^2/g} \qquad \text{8-15}$$

with v_m in cm³(STP)/g. **N_2-only** — do not use it for another adsorbate.

How the measurement is actually done

From slide 8 and the notes:

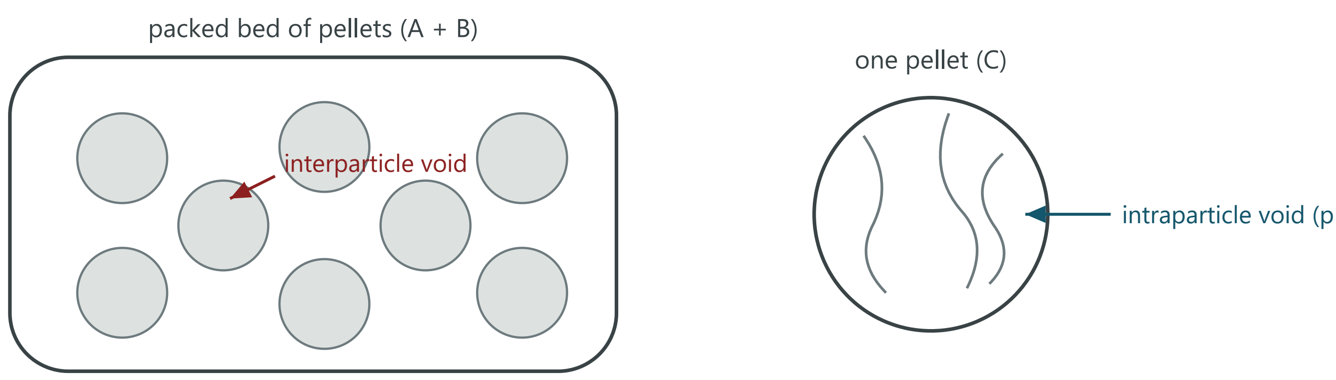
1. **Sample degassing** — clean the surface before anything is measured.
2. A **carrier gas** stream of **N_2 + He** flows over the sample; **N_2 is the adsorbate**, He is the inert carrier.
3. The **test station** measures how much N_2 is taken up at each p/p_0.

Why N_2? The notes give the honest answer: **it is cheaper** — and it physisorbs on anything, which is exactly what you want when measuring total area.

§ 15 · Slide 9

The density set and porosity

Five densities, and they differ only in **which voids the denominator counts**. Get that straight and the whole section is free marks.



CLASS NOTES PP. 20–21 AND SLIDE 9

Density	Numerator	Denominator — the part that matters
True density (ρ_s)	Mass of solid	**Actual volume, excluding all voids**
Skeletal density	Mass of solid	Volume excluding **interparticle voids and open pores**, but **including closed pores**
Bulk density (ρ_B)	Mass of solid	**Loosely packed** volume
TAP density (transverse axial pressure)	Mass of solid	**Densely packed** volume
Pellet density (ρ_p)	Mass of **pellet**	Volume of the **pellet** — pores included

$$\varepsilon_p = \frac{\text{void volume of pellet}}{\text{total volume of pellet}}$$

Pellet porosity — the fraction of the pellet that is pore.

ORDERING CHECK

$\rho_{true} > \rho_{skeletal} > \rho_{pellet} > \rho_{TAP} > \rho_{bulk}$. Each step down adds another class of void to the denominator. If your numbers come out in a different order, you have mislabelled something.

JMS 8-6 — POROSITY FROM MEASUREMENTS YOU CAN ACTUALLY MAKE

$\varepsilon_p = V_g\rho_s / (V_g\rho_s + 1)$ (8-16) from the **helium–mercury** experiment, or simply $\varepsilon_p = V_g\rho_p$ (8-17) if you have the pellet density. He fills the pores (giving solid volume); Hg does not enter them at atmospheric pressure (giving pellet volume). Typical $\varepsilon_p \approx$ **0.5**; a fixed bed is roughly 30% pore / 30% solid / 40% interparticle.

§ 16 · Slide 10

Pore volume — mercury intrusion

Mercury does not wet a catalyst (contact angle ≈ 140°, so cos θ is negative). It must therefore be **forced** into a pore, and the pressure needed tells you the pore radius: **the smaller the pore, the higher the pressure**.

$$a = -\frac{2\,\sigma\cos\theta}{p}$$

Ritter & Drake

a = pore radius · σ = surface tension of mercury · $\theta \approx 140°$ · p = applied pressure

*Sweep the pressure, record the volume of Hg forced in, and convert each pressure to a pore radius — that gives the **pore-volume distribution**, not just the total.*

JMS 8-7 — THE WORKING FORMS

- With σ and θ ≈ 140° substituted: $\boldsymbol{a}$(Å) = 8.75 × 10^5 / $\boldsymbol{p}$(psi) (8-19)
- **Mercury reaches macropores only.** For **micropores**, use N_2 **desorption** and the **Kelvin equation**, with contact angle θ = **0°** for nitrogen (it wets).
- Average pore radius, monodisperse pores only: **ā = $2V_g/S_g$** (8-26) — the Wheeler model. Derive it by treating the pore volume as n straight cylinders of length L: $V_g = n\pi a^2 L$, $S_g = 2n\pi a L$, and divide.

THE THREE CHARACTERISATION EXPERIMENTS, SIDE BY SIDE

Quantity	Method	Physics
Surface area S_g	N_2 physisorption, BET	Multilayer physisorption
Pore volume V_g (macro)	Mercury intrusion	Non-wetting liquid forced in
Pore volume (micro/meso)	N_2 desorption	Capillary condensation, Kelvin equation
Active sites	H_2/CO chemisorption	Chemical bonding — only real sites count

§ 17 · Slides 2–3

Case study: fluidised catalytic cracking

The deck opens with FCC as the industrial anchor for everything that follows: a **fluidised bed** of fine catalyst, with a reactor and a regenerator coupled by a **pressure balance** and a **thermal balance**.

$$\text{Heavy oil} \xrightarrow{\text{catalyst}} \text{Gas, Gasoline, Light Cycle Oil}$$

$$C_{15}H_{32} \xrightarrow{\text{zeolite}} 2\,C_2H_4 + C_3H_6 + C_8H_{18}$$

TYPICAL OPERATING VALUES WORTH REMEMBERING AS ORDERS OF MAGNITUDE

Parameter	Typical value
Reactor temperature	480 – 550 °C
Regenerator temperature	650 – 715 °C
Pressure	1.5 – 3 atm
Catalyst / oil ratio	5.5 (wt/wt)
Residence time	2 – 5 seconds
Catalyst	Y-zeolite (faujasite), particle size 50 – 100 μm

WHY IT IS THE RIGHT OPENING EXAMPLE

Every theme of Module 1 is visible in it at once: a **50–100 μm** particle chosen to keep diffusion length short (§5, §7); a **zeolite** chosen for its micropore structure and acid sites (§13); **coke deposition** forcing continuous regeneration (poisons — just past the syllabus); and a **2–5 s** residence time that only works because the intrinsic rate is fast enough that transport, not chemistry, is what you engineer around.

§ 18

Formula sheet

EVERYTHING IN SCOPE, IN ONE PLACE

Eq.	Relation	Use it when
Power law	$(-r_A) = k_0 e^{-E_a/RT} C_A{}^n$	Any intrinsic rate
Rate bases	$(-r_V)V = (-r_W)W = (-r_S)S_A$; $r_V = \rho_B r_W$	Converting lab rate ↔ design rate
7-1 / 7-2	$r_p = k_m a_m (C_b - C_s) = k\, a_m C_s$	Steady state: transport rate = reaction rate
7-3	$C_s = k_m C_b/(k_m + k)$	Eliminating the unmeasurable surface concentration
7-4	$r_p = a_m C_b \,/\, (1/k + 1/k_m)$	**Global rate**; resistances in series
Ergun	$\Delta P/L = 150\, \mu v (1-\varepsilon)^2/(d_p{}^2 \varepsilon^3) + 1.75\, \rho v^2 (1-\varepsilon)/(d_p \varepsilon^3)$	Pressure drop through the bed; $\Delta P \propto 1/d_p$
8-1 / 8-2	$r_a = kp(1-\theta)$; $r_d = k'\theta$	Langmuir adsorption / desorption rates
8-3	$\theta = Kp/(1+Kp) = v/v_m$	**Langmuir isotherm**, pressure form
8-8	$p/v = 1/(Kv_m) + p/v_m$	Linearised Langmuir — BET starts here
—	$1/v = [1/(Kv_m)](1/p) + 1/v_m$	The linearisation used in class
Temkin	$\theta = k_1 \ln(k_2 p)$	ΔH_{ads} falls linearly with coverage
Freundlich	$\theta = c(p)^{1/n}$, $n > 1$	ΔH_{ads} falls logarithmically with coverage
8-9	$p/[v(p_0 - p)] = 1/(v_m c) + (c-1)p/(c v_m p_0)$	**BET equation**; plot vs p/p_0
—	$c = \exp[(E_1 - E_L)/RT]$	Meaning of the BET constant
8-10 → 8-12	$I = 1/(v_m c)$, $s = (c-1)/(v_m c)$, $v_m = 1/(I+s)$	Monolayer volume from a BET plot
8-13	$S_g = (N_0 v_m / V_{STP}) \cdot \alpha / m$	Specific surface area from monolayer volume
8-14	$\alpha = 1.09[M/(N_0 \rho)]^{2/3}$	Area occupied by one adsorbed molecule
8-15	$S_g = 4.35 \times 10^4\, v_m$ cm²/g	**N_2 at −195.8 °C only**
8-7	$S_g = 6/(\rho_p d_p)$	External area of non-porous spheres
8-16 / 8-17	$\varepsilon_p = V_g \rho_s/(V_g \rho_s + 1)$; $\varepsilon_p = V_g \rho_p$	Porosity from He–Hg or from pellet density
8-18 / 8-19	$a = -2\sigma\cos\theta/p$; $a(\text{Å}) = 8.75 \times 10^5/p(\text{psi})$	**Mercury intrusion**, $\theta \approx 140°$
8-20	$a - \delta = -2\sigma V_l \cos\theta \,/\, [R_g T \ln(p/p_0)]$	**Kelvin equation**, N_2 desorption; $\theta = 0°$
8-26	$\bar{a} = 2V_g/S_g$	Wheeler average pore radius — monodisperse only

Numbers to carry in your head

Quantity	Value
Catalyst surface area (requirement)	**10^2 – 10^3 m²/g**
Catalyst particle size (typical)	50 μm → 300 μm (FCC: 50–100 μm)
Pore size scale	Ångström, 1 Å = 10^{-10} m
Micro / meso / macro pore	< 2 nm · 2–50 nm · > 50 nm
Heat of **physisorption**	0.5 – 5 kcal/g-mol
Heat of **chemisorption**	5 – 100 kcal/g-mol
Activation energy for physisorption	≤ ~1 kcal/mol
N_2 normal boiling point	−195.8 °C (liquid ρ = 0.808 g/cm³)
α for N_2	16.2 Å²
N_0 · V_{STP}	6.02 × 10^{23} · 22,400 cm³/mol (22.4 L)
Contact angle: mercury · nitrogen	140° · 0°
Typical pellet porosity ε_p	≈ 0.5 (bed ≈ 30% pore / 30% solid / 40% interparticle)

§ 20

Rapid self-check

Answer each one out loud before opening it. On the printed copy the answers are already visible — cover them with your hand.

Q A catalyst is added and conversion at equilibrium does not change. Why not?

It lowers the activation energy of the forward and reverse reactions **by the same amount**, so $K_{eq} = k_f/k_r$ is untouched. ΔH and ΔG are properties of the reactants and products, not of the path.

Q List the seven steps in order and mark which are chemical.

1 external transport of A · 2 pore diffusion of A · **3 adsorption** · **4 surface reaction** · **5 desorption** · 6 pore diffusion of B · 7 external transport of B. Steps 3–5 are chemical and together constitute the **intrinsic rate**; 1, 7 are external physical and 2, 6 internal physical. For a non-porous pellet 2 and 6 disappear.

Q Why does mass-transfer resistance hurt more at high temperature?

k rises exponentially with T while k_m is nearly T-independent. In $1/k + 1/k_m$, raising T drives $1/k \to 0$, so the fixed $1/k_m$ dominates. The system drifts into diffusion control and the **apparent activation energy collapses**.

Q Smaller particles give a higher rate per kg. Why not just grind the catalyst finer?

$\Delta P \propto 1/d_p$ (Ergun). Below some size the compressor duty, the risk of fluidising or crushing the bed, and the loss of mechanical strength cost more than the extra rate is worth. Shaped catalysts — lobes, wagon wheels, monoliths — are the way to raise S_A/V *without* shrinking d_p.

Q Three ways to tell physisorption from chemisorption in one sentence each.

(i) Heat: 0.5–5 vs 5–100 kcal/mol. (ii) Temperature: physisorption is a low-T, reversible phenomenon; chemisorption is high-T and irreversible. (iii) Selectivity of surface: anything physisorbs, only a chemically active surface chemisorbs — which is precisely why physisorption measures **area** and chemisorption measures **sites**.

Q State the Langmuir assumptions and derive θ = Kp/(1+Kp).

Uniform surface, no adsorbate–adsorbate interaction, all sites equally active, ΔH_{ads} independent of θ, monolayer only. Then $kp(1-\theta) = k'\theta \Longrightarrow kp = \theta(k'+kp) \Longrightarrow \theta = kp/(k'+kp)$; divide by k' and set $K = k/k'$.

Q A BET plot has intercept $\boldsymbol{I}$ and slope $\boldsymbol{s}$. Get $\boldsymbol{v_m}$ and $\boldsymbol{c}$.

$I = 1/(v_m c)$ and $s = (c-1)/(v_m c)$. Adding them: $I + s = c/(v_m c) = 1/v_m$, so $\boldsymbol{v_m = 1/(I+s)}$ and then $\boldsymbol{c = s/I + 1}$.

Q Which isotherm type would a purely microporous solid give, and why?

Type I. The micropores fill at low p/p_0 and there is no room for multilayer build-up, so the uptake plateaus — the Langmuir pattern.

Q Why is mercury *forced* into pores while nitrogen enters freely?

Mercury does not wet the solid: θ ≈ 140°, cos θ < 0, so the capillary force opposes entry and must be overcome by pressure — hence $a = -2\sigma\cos\theta/p$, small pores needing high pressure. Nitrogen wets (θ = 0°) and even condenses in the pores by capillary condensation, which is why N_2 desorption is the tool for the small end of the distribution.

Q You are given ρ_{bulk}, ρ_{pellet} and ρ_{true}. What can you compute?

Pellet porosity from ρ_p and ρ_s: $\varepsilon_p = 1 - \rho_p/\rho_s$, and the pore volume per gram $V_g = \varepsilon_p/\rho_p$. The bed void fraction follows from $\varepsilon_B = 1 - \rho_B/\rho_p$ — feed that straight into Ergun.

Q Promoter, inhibitor, poison — distinguish them.

Promoter (e.g. Al_2O_3 or K_2O on Fe for ammonia) is added during manufacture to raise activity, selectivity or life. **Inhibitor** (e.g. halogen on Ag/Al_2O_3 for ethylene oxide) is also added deliberately, but to *suppress* an unwanted reaction and raise selectivity. **Poison** is not added at all — it arrives with the feed or is made by the reaction, during operation.

CL 24303 · CRE-II · Quiz 1 revision. Primary sources: class notes (21 pages) and the lecture deck (`CRE–II_Q1`, 10 slides), covering material taught up to and including 14 August 2026. Boxes marked **JMS** are added from J. M. Smith, *Chemical Engineering Kinetics*, Ch. 7 and Ch. 8-1 → 8-10, and are reference only — everything outside those boxes comes from the lectures.